

PackEdge Logistics Report – Final Project

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# PackEdge Logistics Report

## Part One: Algorithmic Solutions

### Key Delivery Problems

PackEdge Shipping and Logistics currently operates a delivery route that passes through multiple stops in Gainesville, Florida. The current data provide estimated delivery times and distances between locations, but there is no evidence that these routes have been optimized using a formal algorithmic approach. As a result, several potential delivery problems may arise.

The first potential problem is route inefficiency. The current plan suggests a predetermined route rather than a dynamically optimized one. Inefficient routing can increase fuel consumption, vehicle wear, and overall delivery time, thereby raising operating costs over time (Christiansen, 2026). If PackEdge's current route is not the shortest or most efficient path, it could result in additional costs for the company, including increased fuel consumption, longer paid hours for employees, and more customer complaints due to slower deliveries.

A second problem is the impact of traffic congestion on delivery times. The provided schedule assumes consistent travel times between locations; however, traffic conditions often vary throughout the day and week. Christiansen (2026) explains that traffic congestion, road closures, and weather conditions are common route planning challenges that can reduce service reliability. Delays caused by these factors may result in inaccurate estimated times of arrival (ETAs), making it more difficult for customers and businesses to plan around deliveries. Improving route selection can reduce the impact of traffic-related delays, enhance delivery reliability, and increase customer satisfaction.

Finally, scalability becomes an issue as PackEdge takes on additional delivery requests. While the current route may be sufficient for the existing number of deliveries, it will become increasingly inefficient and difficult to maintain as the delivery network grows. Route planning becomes more complex as businesses manage additional stops, delivery windows, vehicle capacity limits, and driver schedules (Christiansen, 2026). Implementing an algorithmic approach to route planning will allow PackEdge to adapt more effectively to future growth while maintaining operational efficiency.

These challenges demonstrate the need for a route optimization strategy that can improve delivery efficiency, maintain accurate delivery windows, and support future expansion.

### Algorithmic Solution Comparison

Several algorithms are viable for analyzing and optimizing route efficiency. For PackEdge, three potential approaches are the Floyd-Warshall algorithm, a greedy algorithm, and breadth-first search (BFS). Each algorithm offers different advantages and disadvantages when applied to route optimization.

| Algorithm                | Time Complexity | Space Complexity | Decision-Making Process                                | Suitability for Route Optimization            | Ease of Implementation |
|--------------------------|-----------------|------------------|--------------------------------------------------------|-----------------------------------------------|------------------------|
| Breadth-First Search     | $O(V + E)$      | $O(V)$           | Explores neighboring vertices level by level           | Poor for weighted routes                      | Easy                   |
| Greedy Algorithm         | $O(E \log V)$   | $O(V)$           | Selects the locally optimal choice                     | Well-suited for shortest-path routing         | Moderate               |
| Floyd-Warshall Algorithm | $O(V^3)$        | $O(V^2)$         | Calculates shortest path between all pairs of vertices | Highly accurate but computationally expensive | Moderate               |

Breadth-first search is an efficient graph traversal algorithm that explores vertices one level at a time. Pavicic (2023) notes that BFS visits neighboring vertices before moving to deeper levels of the graph and is commonly used for shortest-path calculations in unweighted graphs. While BFS performs well when all edges have equal weight, it may not perform well for PackEdge because travel distances between locations differ. Since BFS ignores edge weights, it may choose a route with fewer stops rather than the most efficient route to all stops.

A greedy algorithm makes decisions by selecting the best immediate option at each step. When applied to route optimization, a greedy shortest-path approach prioritizes routes with the lowest cumulative cost. Route optimization algorithms are commonly used to determine the best stop order for multiple delivery locations while reducing travel time, distance, and fuel costs (FieldServicely, 2026). This makes a greedy approach well suited for delivery networks where minimizing travel distance and delivery times is important. Greedy algorithms are generally efficient, scalable, and practical for situations like the PackEdge case.

The Floyd-Warshall algorithm calculates the shortest paths between every pair of vertices within a graph. This provides a complete view of the network and guarantees optimal shortest-path information. However, its time complexity makes it significantly more computationally expensive than other approaches. While this algorithm is robust and useful for comprehensive network analysis, it may perform unnecessary calculations when only a single delivery route needs to be optimized. If computational costs are not a concern and PackEdge requires shortest-path information between all delivery locations, Floyd-Warshall could be a strong alternative. However, when optimizing a single delivery route, the additional computational overhead may outweigh its benefits.

From a stability and robustness perspective, both the greedy and Floyd-Warshall algorithms produce consistent results when route distances remain unchanged. At the same time, BFS is less robust because it does not account for weighted travel costs.

Overall, the greedy algorithm provides the best balance between efficiency, scalability, robustness, and ease of implementation for PackEdge. Unlike BFS, it accounts for weighted distances, and unlike Floyd-Warshall, it avoids the computational expense of calculating shortest paths between every possible pair of locations. For these reasons, the greedy algorithm is the most appropriate choice for PackEdge's route optimization needs.

### Modeling Technique

The most appropriate modeling technique for PackEdge is a weighted directed graph. In graph theory, a graph consists of vertices and edges. In a network of delivery locations, the vertices would represent the delivery locations, while the edges would represent the roads or routes between them. Since the provided documentation includes distances between locations, each edge can be assigned a numerical weight corresponding to travel distances.

A directed graph is appropriate because deliveries follow a specific route, and travel conditions may cause travel costs to vary with the direction of travel. A weighted graph allows route optimization algorithms to consider the cost of each connection rather than treating them all as equal. According to Pavicic (2023), weighted graphs assign numerical values to relationships that represent factors such as distance, cost, or travel effort, making them particularly useful for route planning and network optimization. This provides a more realistic representation of the delivery network and supports shortest-path calculations.

Using a weighted directed graph also aligns well with PackEdge's objectives. The model provides a clear representation of delivery locations and their relationships, while also supporting algorithmic analysis through graph traversal and shortest-path algorithms. As a result, it provides an effective foundation for evaluating route optimization strategies and improving delivery efficiency.

### Route Optimization Strategies

Using the weighted directed graph model, several route optimization strategies were considered for PackEdge Shipping and Logistics. Breadth-first search, the Floyd-Warshall algorithm, and a greedy shortest-path approach each provide distinct methods for optimizing the delivery network.

Breadth-first search is less effective for this application because it does not account for weighted travel distances. While it is computationally efficient and easy to implement, it may not produce the most efficient delivery routes. Floyd-Warshall provides optimal shortest-path information between all pairs of locations and offers a comprehensive view of the network. However, its higher computational and memory requirements make it less practical for optimizing a single delivery route.

The greedy shortest-path strategy provides the best balance of efficiency, scalability, robustness, and ease of implementation. By selecting routes with the lowest cumulative travel cost, the strategy can effectively reduce delivery distances while maintaining reasonable computational requirements. Pavicic (2023) notes that weighted graph algorithms account for the costs of traversing edges between vertices, making them particularly useful for route planning and shortest-path problems.

Based on this analysis, the greedy shortest-path strategy was selected as the preferred route optimization strategy for PackEdge Shipping and Logistics.

### Rationale

It is important to analyze multiple route optimization strategies because different algorithms are designed to solve different types of problems. While all three algorithms operate on a graph structure, each uses a different method to navigate and analyze a network. Comparing multiple strategies helps ensure that the selected solution aligns with PackEdge's goals of improving delivery efficiency, maintaining accurate delivery windows, and supporting future growth.

By evaluating multiple approaches instead of immediately selecting a solution, PackEdge can make a more informed decision based on both operational needs and computational efficiency.

Breadth-first search is useful for exploring connections between locations within a graph. In discrete structures, it demonstrates how graph traversal algorithms can be used to explore a network and identify reachable locations. However, because BFS does not consider weighted travel distances, it is not the most effective option for optimizing delivery routes.

The Floyd-Warshall algorithm is useful because it calculates the shortest paths between every pair of locations in the network. This provides a complete picture of the delivery system and guarantees optimal shortest-path information. Within discrete structures, it demonstrates how graph theory and matrix operations can be used to solve complex routing problems. However, the additional computational resources required by the algorithm may not be justified when optimizing a single delivery route.

The greedy shortest-path strategy uses weighted graph relationships to make routing decisions based on the lowest cumulative travel cost (FieldServicely, 2026). This approach directly supports PackEdge’s objective of reducing travel distances while maintaining efficient delivery operations. In addition, the strategy is scalable, practical, and relatively easy to implement compared to more computationally intensive alternatives.

Although the greedy strategy was selected, there are still factors that can affect route performance. Traffic congestion, road construction, accidents, weather conditions, and changing delivery demands can all influence travel times and route efficiency. While no routing strategy can perfectly account for every real-world situation, the greedy shortest-path approach provides an effective balance between accuracy, computational efficiency, and ease of implementation. For these reasons, it was selected as the preferred route optimization strategy for PackEdge Shipping and Logistics.

## **Part Two: Graphical Representation**

### Delivery Locations and Road Connections Sets

The PackEdge delivery network can be represented as a weighted directed graph  $G = (V, E)$ , where  $V$  denotes the set of delivery locations and  $E$  denotes the set of road connections between them.

The set of delivery locations is defined as:

$$V = \{W, C, D1, WS, S, WM1, N8, NN, M, D2, H, VA, WM2, P, UF, RC, KC\}$$

where each element represents a delivery location or the PackEdge warehouse.

### *Delivery Location Vertices Table*



| Symbol | Location                              |
|--------|---------------------------------------|
| W      | PackEdge Warehouse                    |
| C      | Chun Ching Market                     |
| D1     | Dollar Tree (NW 23rd Street)          |
| WS     | Ward's Supermarket                    |
| S      | Satchel's Pizza                       |
| WM1    | Walmart Supercenter (NE 12th Avenue)  |
| N8     | Northeast 8th Avenue                  |
| NN     | Northeast Neighbors                   |
| M      | Magnolia Plantation Bed and Breakfast |
| D2     | Dollar Tree (South Main Street)       |
| H      | Hotel Eleo                            |
| VA     | VA Hospital                           |
| WM2    | Walmart Supercenter (SW 42nd Street)  |
| P      | Publix Westgate Shopping Center       |
| UF     | University of Florida                 |
| RC     | Raising Cane's                        |
| KC     | Karma Cream                           |

The set of road connections is defined as:

$$E = \{(u, v) | u, v \in V, u \neq v\}$$

where each ordered pair  $(u, v)$  represents a possible route between two locations in the delivery network. Each connection is assigned a numerical weight corresponding to the travel distance between locations.

Using separate sets for delivery locations and road connections enables the network to be analyzed with graph-theoretic techniques such as graph traversal, shortest-path calculations, and route optimization algorithms. This approach enables PackEdge to evaluate alternative delivery routes rather than being limited to the current delivery sequence.

### Relations Between Identified Sets

The delivery location set  $V$  contains 17 vertices, including the PackEdge warehouse and 16 delivery locations. In a fully connected directed graph, each location could potentially connect to every other location. Therefore, the complete delivery network could contain as many as  $|V|(|V| - 1) = 17(16) = 272$  possible directed road connections.

While a fully connected graph would provide the most comprehensive representation of the delivery network, displaying all 272 potential connections would significantly reduce readability and make analysis difficult within the scope of this report. For this reason, the graphical representation focuses on the delivery sequence provided in the PackEdge data. This allows the network to be clearly visualized while still demonstrating the relationships among delivery locations, travel distances, and estimated arrival times.

The vertices in the graph represent delivery locations, while the directed edges represent the travel path between consecutive stops. Edge labels indicate the distance traveled between locations, and each vertex includes its corresponding estimated time of arrival (ETA). The directed nature of the graph illustrates the flow of deliveries throughout the route, while the ETA values represent delivery scheduling requirements and time constraints.

By representing the delivery network as a weighted directed graph, the relationships between locations can be analyzed using graph traversal and shortest-path algorithms. Weighted

graphs are commonly used for route planning and network optimization because they allow costs such as distance, time, or travel effort to be associated with connections between locations (Pavicic, 2023). This graphical model provides a foundation for evaluating route efficiency, identifying potential improvements, and supporting route optimization decisions for PackEdge Shipping and Logistics.

The graphical representation focuses on PackEdge's current delivery route. Although the complete delivery network could contain substantially more connections between locations, displaying only the existing route improves readability while still illustrating the relationships between delivery locations, travel distances, and delivery deadlines.

### *Current Delivery Route Representation*

|          |                                                       |
|----------|-------------------------------------------------------|
| 0.000 mi | PackEdge Warehouse<br>Start: 8:00 AM                  |
| 0.825 mi | Chun Ching Market<br>ETA: 8:02 AM                     |
| 4.749 mi | Dollar Tree - NW 23rd St<br>ETA: 8:17 AM              |
| 3.850 mi | Ward's Supermarket<br>ETA: 8:29 AM                    |
| 1.675 mi | Satchel's Pizza<br>ETA: 8:37 AM                       |
| 1.028 mi | Walmart Supercenter - NE 12th Ave<br>ETA: 8:44 AM     |
| 0.833 mi | Northeast 8th Avenue<br>ETA: 8:50 AM                  |
| 0.657 mi | Northeast Neighbors<br>ETA: 8:55 AM                   |
| 1.155 mi | Magnolia Plantation Bed and Breakfast<br>ETA: 9:03 AM |
| 1.501 mi | Dollar Tree - South Main St<br>ETA: 9:11 AM           |
| 0.865 mi | Hotel Eleo<br>ETA: 9:17 AM                            |
| 0.220 mi | VA Hospital<br>ETA: 9:22 AM                           |
| 3.736 mi | Walmart Supercenter - SW 42nd St<br>ETA: 9:35 AM      |
| 3.106 mi | Publix Westgte Shopping Center<br>ETA: 9:47 AM        |
| 1.593 mi | University of Florida<br>ETA: 9:58 AM                 |
| 1.405 mi | Raising Cane's<br>ETA: 10:09 AM                       |
| 0.838 mi | Karma Cream<br>ETA: 10:15 AM                          |

## Road Network Adjacency Matrix

|     | W | C     | D1    | WS    | S     | WM1   | N8    | NN    | M     | D2    | H     | VA    | WM2   | P     | UF    | RC    | KC    |
|-----|---|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| W   | 0 | 0.825 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| C   | 0 | 0     | 4.749 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| D1  | 0 | 0     | 0     | 3.850 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| WS  | 0 | 0     | 0     | 0     | 1.675 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| S   | 0 | 0     | 0     | 0     | 0     | 1.028 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| WM1 | 0 | 0     | 0     | 0     | 0     | 0     | 0.833 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| N8  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0.657 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| NN  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1.155 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| M   | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1.501 | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| D2  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0.865 | 0     | 0     | 0     | 0     | 0     | 0     |
| H   | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0.220 | 0     | 0     | 0     | 0     | 0     |
| VA  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3.736 | 0     | 0     | 0     | 0     |
| WM2 | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3.106 | 0     | 0     | 0     |
| P   | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1.593 | 0     | 0     |
| UF  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1.405 | 0     |
| RC  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0.838 |
| KC  | 0 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

**Vertex Key:** W = Warehouse, C = Chun Ching, D1 = Dollar Tree (NW 23rd St), WS = Ward's Supermarket, S = Satchel's Pizza, WM1 = Walmart (NE 12th Ave), N8 = Northeast 8th Avenue, NN = Northeast Neighbors,

M = Magnolia Plantation, D2 = Dollar Tree (South Main St), H = Hotel Eleo, VA = VA Hospital, WM2 = Walmart (SW 42nd St), P = Publix Westgate, UF = University of Florida, RC = Raising Cane's,

KC = Karma Cream.

The adjacency matrix represents the weighted directed graph of the PackEdge delivery network. Each row represents the starting location, and each column represents the destination location. A nonzero value indicates a direct road connection between two locations, while a value of 0 indicates that no direct connection is represented in the current delivery route.

Because the graph is directed, the matrix is not symmetric. Distances are recorded only in the direction of travel shown in the delivery sequence. The matrix provides a compact representation of the network and can be used by graph traversal and shortest-path algorithms to analyze route efficiency and connectivity.

Since the complete road network was not provided in the PackEdge data, the weighted adjacency matrix represents the current delivery route shown in the graphical representation.

Nonzero values indicate direct travel distances between consecutive delivery locations.

### Algorithmic Analysis

The delivery route representation and adjacency matrix provide two different ways of modeling the PackEdge delivery network. The route representation offers a visual view of the delivery sequence, making it easy to identify the order of stops, travel distances between locations, and estimated arrival times. The directed flow of the route illustrates how deliveries move through the network, while the distance labels and ETAs provide information about travel costs and scheduling requirements.

The adjacency matrix provides a numerical representation of the same weighted directed graph. Graph representations, such as adjacency matrices, enable computational analysis of graph structures and support graph traversal and shortest-path algorithms (Pavicic, 2023). Each row represents a starting location, and each column represents a destination location. Nonzero values indicate direct travel distances between connected locations, while zero values indicate that no direct connection is represented in the current delivery route. This representation organizes the network into a format that can be analyzed mathematically and processed by routing algorithms.

Although the route representation and adjacency matrix display the network differently, both contain the same underlying information. The route representation emphasizes readability and delivery flow, while the adjacency matrix emphasizes mathematical analysis and computation. Together, they provide a complete view of the delivery network and support the evaluation of route efficiency and connectivity.

The selected greedy shortest-path strategy can use the weighted connections represented in both models to evaluate travel costs between locations. By considering the distances of each route segment, the algorithm can identify efficient delivery paths while keeping computational

requirements reasonable. This allows PackEdge to analyze and optimize delivery operations using graph-theory principles and discrete structures.

### Algorithmic Solution

Based on the analysis of multiple route optimization strategies, the greedy shortest-path algorithm was selected as the algorithmic solution for the PackEdge delivery network. This algorithm was chosen because it provides an effective balance between computational efficiency, scalability, robustness, and ease of implementation while accounting for weighted travel distances between delivery locations.

Within the weighted directed graph model, the greedy algorithm evaluates the available connections between locations and prioritizes routes with the lowest cumulative travel cost. This allows the delivery network to be analyzed efficiently while avoiding the additional computational overhead of more complex algorithms such as the Floyd-Warshall algorithm.

The selected algorithm supports PackEdge's goals of improving route efficiency, maintaining accurate delivery windows, and adapting to future growth. By using weighted distances to inform routing decisions, the greedy shortest-path approach provides a practical solution for optimizing delivery operations within the current network.

### Shortest Paths and Connectivity

The adjacency matrix and route representation demonstrate that the PackEdge delivery network forms a connected path from the warehouse to the final delivery location. Because each location is linked to the next location in the delivery sequence, every delivery stop is reachable from the starting location. This connectivity is important because it ensures that deliveries can be completed without encountering isolated or unreachable locations.

Using the weighted distances provided in the network, the shortest path from the PackEdge warehouse to the destination, Karma Cream, follows the delivery sequence represented in the graph and adjacency matrix:

$$W \rightarrow C \rightarrow D1 \rightarrow WS \rightarrow S \rightarrow WM1 \rightarrow N8 \rightarrow NN \rightarrow M \rightarrow D2 \rightarrow H \rightarrow VA \rightarrow WM2 \rightarrow P \rightarrow UF \\ \rightarrow RC \rightarrow KC$$

The total travel distance for this path is: 28.036 *miles*

Because the PackEdge dataset only provides distances between consecutive delivery locations, the shortest-path analysis is limited to the route represented in the graph and adjacency matrix. A more comprehensive shortest-path analysis would require distance information between all delivery locations.

The network's connectivity enables route optimization algorithms to evaluate travel costs between locations and identify efficient delivery paths. In this case, the weighted directed graph provides the information necessary for the greedy shortest-path strategy to analyze route efficiency and support routing decisions. Shortest-path problems are commonly used in transportation networks, including routing systems that must account for travel distance, processing time, and memory requirements (Mouli et al., 2010). By representing locations and travel distances mathematically, the network can be evaluated using graph-theory concepts and matrix operations to improve delivery planning and overall operational efficiency.

### **Part Three: Testing and Optimizing Using Benchmarking**

#### Test Cases

To evaluate the greedy shortest-path strategy, three test cases were created. These test cases examine how the algorithm performs under normal conditions, when a connection is



unavailable, and when the delivery network expands. Each test case uses the weighted directed graph model developed in Part Two.

The first test case is the current PackEdge delivery route. This represents the normal input case because it uses the provided delivery sequence, distances, and estimated arrival times. In this case, all delivery locations are connected in sequence, and the route begins at the PackEdge warehouse and ends at Karma Cream. The total distance for this route is 28.036 miles. This test case confirms that the greedy shortest-path strategy can process the existing delivery network and evaluate the weighted connections between delivery stops.

The second test case represents an edge case in which a route connection becomes unavailable. For example, if the connection between the VA Hospital and the Walmart Supercenter on SW 42nd Street is blocked by road construction or traffic, the current route is disrupted. This tests whether the algorithm can identify that the delivery network is no longer fully connected under the current route structure. In a real delivery system, this would require the algorithm to search for an alternate available connection before continuing the route.

The third test case represents a scalability scenario where PackEdge adds one additional delivery location to the network. This increases the number of vertices and the number of possible road connections in the graph. The purpose of this test case is to determine whether the greedy shortest-path strategy can still evaluate the network efficiently as the number of delivery stops increases. Since the greedy approach avoids computing shortest paths for every possible pair of locations, it remains more practical than the Floyd-Warshall algorithm as the delivery network grows.

Together, these test cases evaluate the selected algorithm under normal, disrupted, and expanded network conditions. The normal case tests basic functionality, the blocked-route case tests connectivity and error handling, and the additional-location case tests scalability. This provides a stronger benchmark than testing only the original delivery route.

### Evaluate

To evaluate the selected greedy shortest-path strategy, the algorithm was applied to each of the three test cases developed for the PackEdge delivery network. The pseudocode was executed conceptually using the weighted directed graph and adjacency matrix developed in Part Two. The results were then recorded and compared to evaluate the functionality and effectiveness of the selected algorithm.

#### *Greedy Shortest-Path Pseudocode:*

GreedyShortestPath(Graph G)

current = Warehouse

while current is not final destination

    select connected vertex with lowest travel cost

    move to selected vertex

    record distance traveled

return completed route

*Test Case 1: Existing Delivery Route:*

| Step | Current Location            | Next Location Selected      | Distance (mi) |
|------|-----------------------------|-----------------------------|---------------|
| 1    | Warehouse                   | Chun Ching Market           | 0.825         |
| 2    | Chun Ching Market           | Dollar Tree (NW 23rd St)    | 4.749         |
| 3    | Dollar Tree (NW 23rd St)    | Ward's Supermarket          | 3.850         |
| 4    | Ward's Supermarket          | Satchel's Pizza             | 1.675         |
| 5    | Satchel's Pizza             | Walmart (NE 12th Ave)       | 1.028         |
| 6    | Walmart (NE 12th Ave)       | Northeast 8th Avenue        | 0.833         |
| 7    | Northeast 8th Avenue        | Northeast Neighbors         | 0.657         |
| 8    | Northeast Neighbors         | Magnolia Plantation         | 1.155         |
| 9    | Magnolia Plantation         | Dollar Tree (South Main St) | 1.501         |
| 10   | Dollar Tree (South Main St) | Hotel Eleo                  | 0.865         |
| 11   | Hotel Eleo                  | VA Hospital                 | 0.220         |
| 12   | VA Hospital                 | Walmart (SW 42nd St)        | 3.736         |
| 13   | Walmart (SW 42nd St)        | Publix Westgate             | 3.106         |
| 14   | Publix Westgate             | University of Florida       | 1.593         |
| 15   | University of Florida       | Raising Cane's              | 1.405         |
| 16   | Raising Cane's              | Karma Cream                 | 0.838         |

**Result:** The algorithm successfully evaluated the route from the PackEdge warehouse to Karma Cream. All delivery locations remained connected, and the route was completed successfully. The total route distance was 28.036 miles.

*Test Case 2: Blocked Route Connection:*

The connection between VA Hospital and Walmart Supercenter (SW 42nd Street) is removed due to road construction.

| Step  | Current Location              | Result                        |
|-------|-------------------------------|-------------------------------|
| 1-11  | Warehouse through VA Hospital | Route proceeds normally       |
| 12    | VA Hospital                   | No available connection found |
| Final | Route terminated              | Destination unreachable       |

**Result:** The algorithm successfully identified that the delivery route could not continue once the connection between VA Hospital and Walmart Supercenter was removed. The graph became disconnected, preventing the route from reaching the remaining delivery locations.

*Test Case 3: Additional Delivery Location:*

A new delivery location is added between Publix Westgate Shopping Center and the University of Florida.

| Step  | Current Location                  | Result                         |
|-------|-----------------------------------|--------------------------------|
| 1-13  | Warehouse through Publix Westgate | Route proceeds normally        |
| 14    | Publix Westgate                   | New delivery location selected |
| 15    | New delivery location             | University of Florida selected |
| Final | Route completed                   | Destination reached            |

**Result:** The algorithm successfully incorporated the additional delivery location into the network. Although the graph contained an additional vertex and additional route calculations, the route remained connected and was completed successfully.

The results demonstrate that the greedy shortest-path strategy functions effectively under a variety of operating conditions. In the normal operating scenario, the algorithm successfully evaluated the existing delivery route and completed all deliveries. In the blocked-route scenario, the algorithm correctly identified a connectivity issue that prevented the route from completing.

In the expanded-network scenario, the algorithm continued to perform efficiently despite the addition of a new delivery location.

These results indicate that the greedy shortest-path strategy provides an effective balance between route efficiency, scalability, and computational performance. The algorithm successfully analyzes weighted route costs, identifies connectivity issues, and adapts to network growth, making it a practical solution for PackEdge Shipping and Logistics.

#### Algorithmic Optimization:

Although the greedy shortest-path strategy provides an effective solution for the current PackEdge delivery network, several opportunities exist for future optimization. One opportunity is to incorporate real-time traffic data into the routing process. The current model uses static travel distances and estimated arrival times, which may not accurately reflect changing road conditions throughout the day. Integrating real-time traffic information could improve route accuracy and delivery reliability. Modern route optimization systems increasingly incorporate live traffic and roadway data to improve routing decisions and reduce disruptions caused by changing conditions (FieldServicely, 2026).

Another opportunity is the use of dynamic route recalculation. In the current model, route decisions are based on the information available when the route is created. However, unexpected events such as road construction, accidents, or weather conditions may affect travel times after deliveries have begun. A dynamic routing system could adjust the delivery sequence when disruptions occur, helping maintain efficient operations. This capability would allow PackEdge to respond more effectively to changing conditions and maintain reliable delivery schedules.

As the delivery network grows, PackEdge may also benefit from more advanced route optimization techniques. While the greedy shortest-path strategy remains practical and computationally efficient for the current network, larger delivery systems may require additional optimization methods to evaluate more delivery locations and route alternatives.

Finally, collecting additional distance and travel-time data between delivery locations would enable a more comprehensive graph representation of the delivery network. The current analysis is limited to the route information provided by the PackEdge Shipping and Logistics Findings document. A more complete set of distance and travel-time data would enable more detailed route comparisons and support more advanced optimization strategies.

#### Final Justification

The greedy shortest-path strategy was selected because it provides an effective balance between efficiency, scalability, robustness, and ease of implementation. The PackEdge delivery network consists of multiple delivery locations connected by weighted travel distances. By evaluating the travel cost of each connection, the greedy strategy can identify efficient delivery paths while avoiding the computational overhead of more complex algorithms. Compared to breadth-first search, the greedy approach accounts for weighted travel distances. Compared to the Floyd-Warshall algorithm, it requires fewer computational resources while still producing practical routing decisions.

The implemented strategy addresses the primary problems identified in the PackEdge delivery network. Route inefficiencies are reduced by evaluating weighted travel distances rather than relying solely on a fixed delivery sequence. The strategy also supports future growth by

remaining practical as additional delivery locations are added to the network. Furthermore, the graph-based model enables the identification and evaluation of disruptions, such as blocked routes or unavailable connections, improving overall route planning and delivery reliability.

The current network requires efficient routing, reliable delivery scheduling, and adaptability to future expansion. The weighted directed graph model satisfies these needs by representing delivery locations as vertices and travel distances as weighted edges. This structure provides a clear representation of the delivery network while supporting graph traversal, shortest-path analysis, and route optimization techniques. As the delivery network grows, the selected strategy can continue to support route evaluation without requiring excessive computational resources.

Network performance can be measured and monitored through several key performance indicators. These include total route distance, delivery completion time, fuel consumption, on-time delivery percentage, and customer satisfaction metrics. Comparing these measurements before and after route optimization would allow PackEdge to evaluate the effectiveness of the selected strategy. Regular monitoring of these metrics would also help identify opportunities for future improvements and operational adjustments.

Future requirements for network users and applications may include support for additional delivery locations, real-time traffic information, dynamic route recalculation, and increased delivery volume. As delivery networks become larger and more complex, route optimization systems increasingly rely on real-time data integration and dynamic routing capabilities to maintain efficiency and delivery reliability (FieldServicely, 2026). As customer demand grows, PackEdge may also require integration with GPS systems, traffic monitoring

services, and automated route-planning tools. The weighted directed graph model provides a flexible foundation that can accommodate these future requirements while continuing to support efficient delivery operations.



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